

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency ω : rad/s; f : Hz

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity m/s ; *nonmagnetic*: $\mu_r = 1$

$$u_p = \frac{c}{\sqrt{\epsilon_r}} \qquad c = 3 \times 10^8 \text{ m/s}$$

Wavenumber k rad/m

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$$

Wavelength & impedance m & Ω

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \qquad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \text{ }\Omega$$

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. *Base*: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0 \cos(\omega t - k\hat{k} \cdot \mathbf{r} + \phi_0)\hat{e}$$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
 k = wavenumber (rad/m): how fast phase changes in space;
 $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$.
 ϕ_0 = initial phase constant (rad): shift set by given condition at known t, position.
 $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$.
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$.
H from $\mathbf{E}$ $\hat{k}$, $\hat{E}$, $\hat{H}$ form right-hand set

$$\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$$

Find ϕ_0 given $E = E_1$ at $t = 0$, position r_0

$$E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$$

$\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

LOSSLESS vs. LOSSY — E & H

	Lossless ($\sigma = 0$)	Lossy ($\sigma \neq 0$)
Impedance	$\eta = \sqrt{\mu/\epsilon}$	$\eta_c = \eta_c e^{j\theta_\eta}$
Prop.	e^{-jkz}	$e^{-\alpha z}e^{-j\beta z}$
$\mathbf{E}(t)$	$\text{Re}\{\tilde{\mathbf{E}}e^{j\omega t}\}$	<i>(same)</i>
$\mathbf{H}(t)$	$\text{Re}\{\tilde{\mathbf{H}}e^{j\omega t}\}$	<i>(same)</i>
$\tilde{\mathbf{H}}$ from $\tilde{\mathbf{E}}$	$\frac{1}{\eta}(\hat{k} \times \tilde{\mathbf{E}})$	$\frac{1}{\eta_c}(\hat{k} \times \tilde{\mathbf{E}})$
$\tilde{\mathbf{E}}$ from $\tilde{\mathbf{H}}$	$-\eta(\hat{k} \times \tilde{\mathbf{H}})$	$-\eta_c(\hat{k} \times \tilde{\mathbf{H}})$
Phase	in phase	H lags E by θ_η

Lossless shortcuts (wave in +z, E along $\hat{e}$):

$\mathbf{E}(z, t)$	$E_0 \cos(\omega t - kz + \phi)\hat{e}$	n/a
$\mathbf{H}(z, t)$	$\frac{E_0}{\eta} \cos(\omega t - kz + \phi)(\hat{k} \times \hat{e})$	n/a
$\mathbf{H}(t)$ direct	$\frac{1}{\eta}(\hat{k} \times \mathbf{E}(t))$	n/a
$\mathbf{E}(t)$ direct	$-\eta(\hat{k} \times \mathbf{H}(t))$	n/a

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k-axis $\Rightarrow$ + direction. Plus $\Rightarrow$ - direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

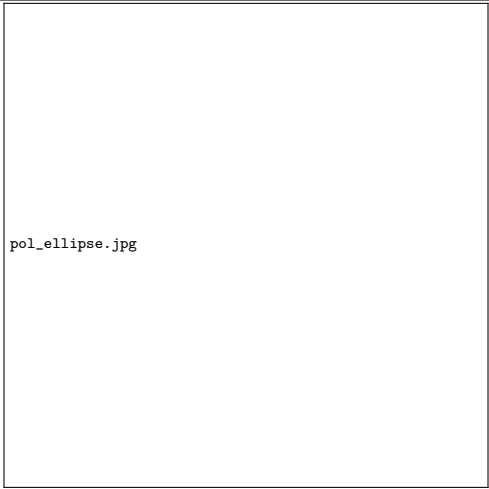
$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$

Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS ($\theta \equiv \omega t - kz$)

Input	Result	Input	Result
$\sin(\theta + \pi/2)$	$\cos \theta$	$\cos(\theta + \pi/2)$	$-\sin \theta$
$\sin(\theta - \pi/2)$	$-\cos \theta$	$\cos(\theta - \pi/2)$	$\sin \theta$
$\sin(\theta \pm \pi)$	$-\sin \theta$	$\cos(\theta \pm \pi)$	$-\cos \theta$

POLARIZATION — GENERAL WAVE



$$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$$

a_x, a_y : bounding box half-widths (V/m); δ : phase diff (rad)
 $\delta = 0$ or $\pi \Rightarrow$ linear pol. ($\chi = 0$, no ellipse)
Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$
Rotation angle γ (major axis from $\hat{x}$) $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0) \cos \delta$$

If $\text{sign}(\gamma) \neq \text{sign}(\delta)$: $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.
Ellipticity angle χ $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0) \sin \delta$$

$$\text{Axial ratio } R = 1/|\tan \chi| \text{ major axis} = R \times \text{minor}; R \geq 1$$

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.		
χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0) \cos \delta, \quad D = \cos(2\psi_0)$:			
N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	-	$180^\circ - \arctan N/D $	$\pi - \arctan N/D $
-	-	$180^\circ + \arctan N/D $	$\pi + \arctan N/D $
-	+	$360^\circ - \arctan N/D $	$2\pi - \arctan N/D $

$\gamma = (2\gamma)/2. \quad D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.
Convert: $\theta_{rad} = \theta_o \times \frac{\pi}{180}$; $\theta_o = \theta_{rad} \times \frac{180}{\pi}$.

PHASOR $\leftrightarrow$ TIME DOMAIN

Phasor coeff	Time domain (add $e^{-\alpha z}$)
A (real)	$A \cos(\omega t - \beta z)$
$-jA$	$A \sin(\omega t - \beta z)$
$+jA$	$-A \sin(\omega t - \beta z)$
$Ae^{j\phi}$	$A \cos(\omega t - \beta z + \phi)$
$A + jB$	$ Z \cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}, \quad \theta = \arctan(B/A)$ (check quadrant)

AVERAGE POWER DENSITY (POYNTING)

General W/m^2	
$\mathbf{S}_{av} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$	
Lossless shortcut $ E_0 ^2 = a_x^2 + a_y^2 + a_z^2$	
$\mathbf{S}_{av} = \frac{ E_0 ^2}{2\eta} \hat{k}$	
Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{av}$ is a negative x-component.	

LOSSY MEDIA — CLASSIFICATION

Loss tangent $\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0}$		
ϵ''/ϵ'	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx
$\epsilon' = \epsilon_r\epsilon_0$; $\epsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\epsilon_r = \mu_r = 1$.		

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\epsilon''/\epsilon')^2}$)

$$\alpha = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X-1} \text{ Np/m}, \quad \beta = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X+1} \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f} \text{ (m)}, \quad u_p = \frac{\omega}{\beta} \text{ (m/s)}$$

$$\eta_c = \frac{\eta}{X^{1/2}} e^{j\theta_\eta} \text{ (}\Omega\text{)}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'} \text{ (rad)}$$

$$\text{Rect: } \eta_c = |\eta_c| \cos \theta_\eta + j|\eta_c| \sin \theta_\eta, \quad |\eta_c| = \eta/X^{1/2}$$
$$\eta = \eta_0/\sqrt{\epsilon_r}; \epsilon' = \epsilon_r\epsilon_0.$$

LOSSLESS ($\sigma = 0$)

$$\alpha = 0 \quad \beta = \omega\sqrt{\mu\epsilon} = k \text{ (rad/m)} \quad \lambda = u_p/f \text{ (m)}$$
$$\eta_c = \eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{, real; } \theta_\eta = 0)$$
$$u_p = 1/\sqrt{\mu\epsilon} = c/\sqrt{\epsilon_r} \text{ (m/s)}$$

LOW-LOSS MEDIUM ($\epsilon''/\epsilon' \ll 1$)

$$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2\eta} \text{ (Np/m)}$$

$$\beta \approx \omega\sqrt{\mu\epsilon'} = \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad u_p \approx \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s)}$$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} = \eta \text{ (}\Omega\text{, } \angle \approx 0)$$

GOOD CONDUCTOR ($\epsilon''/\epsilon' \gg 1$)

$$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma} \text{ (Np/m, rad/m)}$$

$$u_p \approx \sqrt{4\pi f / \mu \sigma} \text{ (m/s)}, \quad \lambda \approx u_p/f \text{ (m)}$$

$$\eta_c \approx (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma} \text{ (}\Omega\text{)}$$

$\tilde{\mathbf{H}}$ PHASOR $\rightarrow \tilde{\mathbf{E}}$ IN LOSSY MEDIA

Step 1 — read α, β

$$\tilde{\mathbf{H}} \propto e^{-\alpha y} e^{-j\beta y} \Rightarrow \gamma = \alpha + j\beta$$

Step 2 — η_c (nonmagnetic: $\mu_r = 1$)

$$\eta_c = \frac{j\omega\mu_0}{\gamma} = \frac{\omega\mu_0(\beta + j\alpha)}{\alpha^2 + \beta^2}$$

Step 3 — find $\tilde{\mathbf{E}}$

$$\tilde{\mathbf{E}} = -\eta_c (\hat{k} \times \tilde{\mathbf{H}})$$

Step 4 — time domain

$$Ae^{j\phi} e^{-\alpha y} \xrightarrow{\text{Re}\{ \cdot e^{j\omega t} \}} Ae^{-\alpha y} \cos(\omega t - \beta y + \phi)$$

Convert each component $A + jB \rightarrow |Z|\angle\theta$ first.

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
 $k = \omega\sqrt{\epsilon_r}/c$ — wavenumber (rad/m)
 $\lambda = 2\pi/k$ — wavelength (m)
 $u_p = c/\sqrt{\epsilon_r}$ — phase vel. (m/s)
 $\eta = \eta_0/\sqrt{\epsilon_r}$ — impedance (Ω)
 $c = 3 \times 10^8$ m/s
 $\eta_0 = 377 \Omega$ (free space)

LOSSLESS MEDIA

$\sigma = 0$ (vacuum, air, ideal dielectric)
 $\epsilon'' = \sigma/\omega = 0 \Rightarrow \epsilon = \epsilon'$ (real)
 $\alpha = 0$; no attenuation
 $\eta = \eta_0/\sqrt{\epsilon_r}$ real (Ω)
 $\eta_0 = 377 \Omega$ (free space)
E and **H** in phase ($\theta_\eta = 0$)

LOSSY MEDIA

α — atten. const (Np/m)
 β — phase const (rad/m)
 $\gamma = \alpha + j\beta$ — prop. const (m^{-1})
 η_c — complex impedance (Ω)
 $\epsilon' = \epsilon_r\epsilon_0$; $\epsilon'' = \sigma/\omega$ (F/m)
 μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
 $\mu = \mu_r\mu_0$; $\mu_0 = 4\pi \times 10^{-7}$ H/m
 $\delta_s = 1/\alpha$ — skin depth (m)

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
 $\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
 a_R, a_L, a_x in V/m; k in rad/m
Linear: $a_R = a_L = a_x/2$
 $-j = e^{-j\pi/2}$: $\cos \rightarrow \sin$
 $+j = e^{+j\pi/2}$: $\cos \rightarrow -\sin$

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
 $\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
 $\arccos(2/3) \approx 48.2^\circ = 0.84$ rad
 $\arctan(4/3) \approx 53.1^\circ = 0.927$ rad
 $\arctan(3/4) \approx 36.9^\circ = 0.644$ rad
 $\arctan(24/7) \approx 73.7^\circ = 1.287$ rad